

KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

arXiv: 2504.03553

Tagline: Agents autonomously regulate when to act, reflect, or retrieve

Problem: 'Flood Irrigation' Knowledge Injection

Current paradigms over-apply training signals regardless of situation:

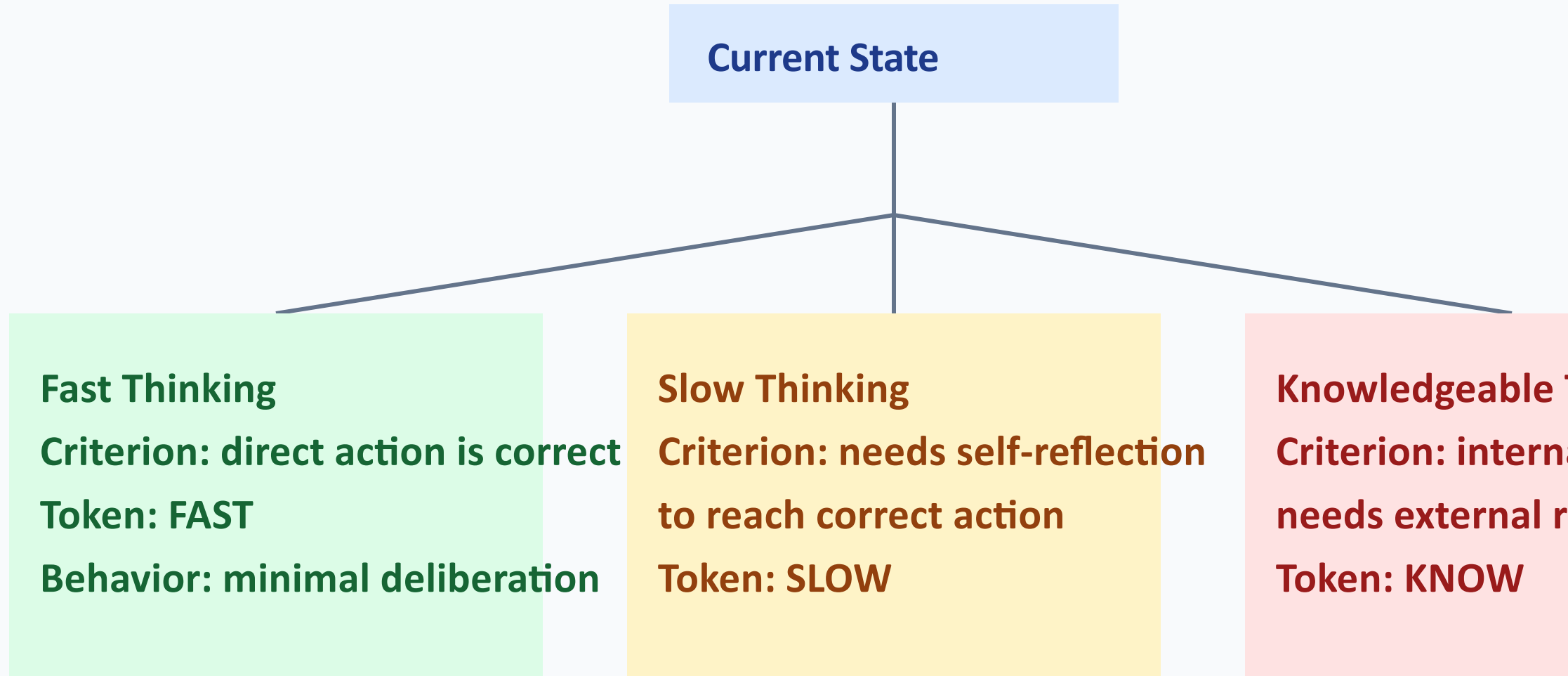
- Trajectory imitation: injects gold paths everywhere
- Trial-and-error reflection: reflects even when unnecessary
- Knowledge-augmented agents: query external knowledge at every step

Observed limitation

**Indiscriminate knowledge use can
and wastes compute/token budget**

Human-like metacognition suggests self
act fast when obvious, reflect when un

Three Situational Thinking Modes



KnowSelf Pipeline

1) Data Construction

Heuristic situation judgment on self-explored trajectories

Attach special tokens:

FAST, SLOW, KNOW

2) Two-Stage Training

Stage A: SFT learns token-conditioned reasoning/acting behavior

Stage B: RPO improves policy quality under situational supervision

3) Self-Aware

Model emits s

FAST→act dire

SLOW→reflect

KNOW→retrie

Qualitative Examples (ALFWorld)

Task A: Put a hot mug in cabinet
KnowSelf uses reflection to heat mug,
then places correctly.
Baseline (OpenAI-O1): fails sequence.

Figure reference: figure_1.jpg (left)

Task B: Put a saltshaker in drawer
KnowSelf triggers knowledge use and
completes placement.
Baseline (DeepSeek-R1): gets stuck

Figure reference: figure_1.jpg (right)

ALFWorld Results: Structured Comparison

Model	Method	Success Rate	Navigation	Object Use	Tool Use	Physical Interaction	Communication	Planning	Overall Score
GPT-4o	ReAct	-	77.41	67.39	69.05	80.95	52.94	80.00	76.12
GPT-4o	Reflexion	-	80.64	67.39	70.48	83.33	52.94	80.00	77.61
GPT-4o	ExpeL	100%	83.87	69.57	73.81	80.95	52.94	80.00	79.85
Llama-8B	ReAct	-	87.10	67.39	66.67	85.71	52.94	84.29	76.87
Llama-8B	Reflexion	-	87.10	73.91	67.62	83.33	55.88	82.86	78.36
Llama-8B	ETO	-	87.10	69.57	67.62	83.33	52.94	82.86	77.61
Llama-8B	KnowAgent	100%	87.10	67.39	72.38	80.95	52.94	84.29	78.36
Llama-8B	WKM	100%	87.10	71.74	70.48	80.95	52.94	84.29	78.36
Llama-8B	KnowSelf	15%	93.55	76.09	77.14	88.10	52.94	86.67	84.33
Gemma-2B	ReAct	-	87.10	58.70	64.76	80.95	52.94	82.86	74.63
Gemma-2B	Reflexion	-	87.10	60.87	66.67	76.19	52.94	81.43	74.63
Gemma-2B	ETO	-	90.32	69.57	66.67	78.57	52.94	77.14	77.61
Gemma-2B	KnowAgent	100%	90.32	63.04	66.67	80.95	52.94	82.86	76.12
Gemma-2B	WKM	100%	87.10	65.22	69.52	80.95	52.94	78.57	76.87
Gemma-2B	KnowSelf	26%	93.55	73.91	76.19	83.33	52.94	79.85	79.85

Efficiency Advantage: Accuracy vs Knowledge Usage

KnowSelf reaches best overall scores with ~15–26% knowledge usage;
full-knowledge baselines use 100% but underperform.

